Sebastian Tamayo

Senior Software Engineer

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Accomplished Senior Web Developer with over a decade of experience architecting, developing, and optimizing scalable, high-performance web applications. Specialized in building intuitive, mobile-first interfaces with Vue.js, and designing secure, extensible backend systems using Node.js, Express.js, and RESTful APIs.

Proficient across the full software development lifecycle—from distributed system architecture and relational/NoSQL data modeling to CI/CD automation, comprehensive testing strategies, and cloud-native deployments on AWS, Docker, and Kubernetes. Highly effective in Agile, cross-functional teams, consistently delivering user-centric, business-aligned solutions through iterative development and close collaboration with stakeholders.

Renowned for writing clean, maintainable, and testable code, mentoring engineering teams, and advocating for best practices based on SOLID principles, Clean Architecture, and Test-Driven Development (TDD). Adept at frontend performance optimization, modular Vue component architecture, and developing scalable, resilient backend services using Node.js, Redis, and asynchronous job queues (e.g., BullMQ, RabbitMQ).

Driven by a passion for innovation and continuous learning, with deep expertise in the JavaScript ecosystem, Vue framework, Node.js microservices, and modern DevOps toolchains. Committed to building impactful, real-world software solutions that deliver superior user experience and measurable business value.

PROFESSIONAL EXPERIENCE

UST

Senior Software Engineer

June 2022 – June 2025 (3 years)

- Collaborated within a cross-functional Agile Scrum team, mentoring junior developers and conducting in-depth code reviews emphasizing SOLID principles, scalable architecture, and clean, testable code across both frontend and backend layers.
- Revamped a high-traffic, customer-facing application by implementing a Vue.js frontend featuring responsive design, WCAG-compliant accessibility, and real-time interactivity. Integrated live streaming via AWS IVS and external services like the Shopify Storefront API, significantly boosting user engagement and conversions.
- Engineered robust backend services using Node.js and Express.js, developing secure, modular RESTful APIs with JWT authentication, TypeORM (or Prisma), and PostgreSQL, ensuring transactional integrity and flexible, scalable data models.
- Designed a service-oriented backend architecture containerized with Docker and deployed via Amazon ECS. Enabled asynchronous operations and real-time communication using BullMQ (Redis queues) and Socket.IO, enhancing overall system responsiveness and scalability.
- Spearheaded the development of an AI-powered chatbot leveraging AWS Lex, AWS Lambda, and DynamoDB, enabling dynamic, context-aware automation and reducing first-response times across customer support channels.
- Optimized frontend performance with Vue.js dynamic imports, lazy loading, component-level memoization, and Vite-based build enhancements. Backend optimization included query tuning, Redis caching, and Node.js cluster mode, resulting in a 30% improvement in application speed and reliability.
- Developed and maintained automated CI/CD pipelines using GitHub Actions and Jenkins, integrating unit, integration, and E2E testing with Jest, Vue Test Utils, and Cypress, significantly reducing deployment time and improving code quality through rapid feedback loops.

Tech Stack: Vue.js, Pinia (Vuex), TypeScript, HTML5, SCSS, Node.js, Express.js, REST API, PostgreSQL, Redis, DynamoDB, WebSockets, Shopify Storefront API, AWS (ECS, Lambda, IVS, Lex, S3, CloudWatch), Docker, GitHub Actions, Jenkins, Jest, Vue Test Utils, Cypress, Webpack, Vite, WCAG, Figma, Postman, SEO, TDD

Gorilla Logic

Full Stack Engineer

February 2018 – May 2022 (4 years 4 months)

- Designed and implemented responsive, mobile-first user interfaces using Vue.js, TypeScript, and SCSS, resulting in a 40% increase in mobile user retention and a 30% boost in overall platform engagement.
- Built scalable, reusable component libraries for enterprise dashboards, utilizing Pinia (or Vuex) for centralized state management, ensuring UI consistency and accelerating delivery across frontend teams.
- Developed secure, high-performance backend services using Node.js and Express.js, architecting clean, modular RESTful APIs while adhering to domain-driven design and object-oriented best practices with TypeORM (or Prisma).
- Enabled real-time, dynamic content updates with Socket.IO and Server-Sent Events (SSE), powering features like live notifications and real-time dashboards with seamless UX.
- Containerized Node.js services using Docker and deployed via AWS ECS, while automating infrastructure provisioning using AWS CloudFormation, which improved deployment consistency and reduced release time.
- Enhanced backend performance through advanced PostgreSQL query tuning, index optimization, and schema normalization, achieving a 15% reduction in API response latency during high-traffic periods.
- Implemented asynchronous task queues with BullMQ (Redis-based) and optionally RabbitMQ, improving scalability, decoupling processes, and enhancing fault tolerance across backend workflows.
- Practiced Test-Driven Development (TDD) using Jest, Vue Test Utils, and integrated a comprehensive observability stack with Prometheus, Grafana, Elastic APM, and AWS CloudWatch for real-time monitoring and rapid incident response.

Tech Stack: Vue.js, Pinia (Vuex), TypeScript, SCSS, Node.js, Express.js, REST API, BullMQ, PostgreSQL, Redis, RabbitMQ, WebSockets (Socket.IO), SSE, Docker, AWS (ECS, Lambda, S3, CloudFormation, CloudWatch), Prometheus, Grafana, Elastic APM, Git, CI/CD, Jest, Vue Test Utils, Postman, Figma, SEO, WCAG, TDD

Sophos Solutions

5 years 6 months

Senior Web Developer

September 2015 – February 2018 (2 years 6 months)

- Spearheaded the development of a high-performance Single Page Application (SPA) for a financial services platform using Vue.js, Composition API, and Pinia (or Vuex), resulting in a 30% increase in user engagement and faster client-side performance.
- Modernized a legacy monolithic architecture by decoupling the frontend and integrating it with scalable, secure RESTful APIs built with Node.js, Express.js, and JWT-based authentication, improving server response time by 40% and supporting maintainable service-oriented architecture.
- Enhanced frontend performance using advanced optimization techniques such as code splitting, lazy loading, tree shaking, and asset compression. Improved SEO by implementing structured metadata, Open Graph, and JSON-LD, which led to a 35% reduction in page load times.
- Delivered fully responsive, mobile-first UIs adhering to WCAG 2.1 accessibility standards and optimized for Google AMP, contributing to a 25% increase in mobile traffic and improved Lighthouse performance scores.
- Built and maintained a robust full-stack testing strategy using Vue Test Utils, Jest, Cypress, Playwright, and Supertest, with CI/CD automation via GitHub Actions and Jenkins, reducing post-release defects by 75%.
- Collaborated closely with UI/UX designers in Figma and Adobe XD to implement a modular, scalable component system for real-time financial dashboards, enhancing dashboard rendering speed by 50% and improving team productivity across parallel workstreams.

Tech Stack: Vue.js, Composition API, Pinia (Vuex), TypeScript, HTML5, SCSS, Node.js, Express.js, REST API, JWT, MySQL, Redis, Vite, Webpack, GitHub Actions, Jenkins, Supertest, Playwright, Jest, Cypress, Vue Test Utils, Postman, WCAG, Google AMP, SEO, Open Graph, JSON-LD, Google Analytics, Adobe XD, Figma, CI/CD, TDD

Web Developer

September 2012 – February 2015 (2 years 6 months)

- Developed responsive, accessible, and standards-compliant web applications using Vue.js, TypeScript, HTML5, and SCSS, ensuring cross-browser consistency, WCAG 2.1 compliance, and improved engagement across desktop and mobile users.
- Engineered reusable and modular Vue components utilizing the Composition API, Vue Router, and Pinia (or Vuex) to create scalable frontend architectures aligned with long-term product growth and design system principles.
- Designed and integrated RESTful APIs using Laravel and Laravel Sanctum, delivering secure and extensible backend services to power complex workflows and structured business logic.
- Optimized development workflows by configuring Vite, Webpack, and Laravel Mix, automating environment-specific builds and streamlining asset bundling for production-ready deployments.
- Collaborated with backend developers to implement caching strategies using Redis and Laravel's Cache and Queue APIs, significantly improving server response times and scalability under high load.
- Tuned MySQL performance through query optimization, index planning, and Eloquent ORM profiling to reduce latency and increase throughput for data-heavy endpoints.
- Reduced frontend load times by 35% using code splitting, lazy loading, image compression, and minification, contributing to improved SEO performance and faster time-to-interactive for users.
- Maintained robust testing coverage using Jest, Laravel Dusk, and PHPUnit, integrated into GitHub Actions pipelines for continuous delivery and post-deployment validation.

Tech Stack: Vue.js, TypeScript, HTML5, SCSS, Composition API, Vue Router, Pinia (Vuex), PHP, Laravel, Laravel Sanctum, REST API, MySQL, Redis, Laravel Mix, Vite, Webpack, GitHub Actions, Jenkins, Azure App Service, Google Analytics, SEO, WCAG, CI/CD, Postman, Jest, Laravel Dusk, PHPUnit, TDD, Figma

EDUCATION

Universidad Católica de Pereira 04.2005 – 09.2012

Bachelor of Computer Science

SKILLS

Languages:	C++, JS(ES6), TS, Python, C#, Rust, PHP, Java, HTML5
Frameworks and Libraries:	Vue, Vuex, React.js, Next.js, Gatesby, Angular, Node.js, Express, React Native, Redux, RxJS, REST API, Django, JavaSpringBoot, Antd, GraphQL, Ably, Laravel, Tailwind CSS, RabbitMQ, Kafka, TypeORM, Shopify, WordPress
DBMS	MySQL, MS SQL, PostgreSQL, MongoDB, AWS DynamoDB, AWS RDS, Azure SQL Database
IDEs and Tools:	Visual Studio Code, Jenkins, Cypress, Jest, Jasmine, Terraform, Docker, Kubernetes, Redux DevTools, Babel, Webpack, Storybook, Azure DevOps, Git Bash, PowerShell
Cloud Services:	AWS Cloud Services, Microsoft Azure, GCP
Other skills:	Problem Solving, Full Stack Development, Critical Thinking, Leadership, Algorithm Design, Team-Oriented, SOLID, Agile (Scrum), Microservices, Test-Driven Development (TDD), Diagnosing, Business Analysts